

Data Science with Python – Module 3 Exercise

Hands-on Exercise No. 3

Course: Data Science with Python

Module: Machine Learning Basics

Total Marks: 10

Due Date: 25/06/2026

Instructions

Prepare your solution in MS Word.

Submit in **.doc** or **.docx** format.

Late or copied submissions will not be accepted.

Learning Outcome

After completing this exercise, you will be able to:

- Understand supervised learning concepts.
 - Build regression and classification models.
 - Split datasets into training and testing sets.
 - Evaluate model performance using standard metrics.
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Problem Statement

You are working as a Junior Data Scientist for a finance company. The client wants to predict customer loan approvals and loan amounts based on historical data. Your task is to build machine learning models using Python and evaluate their performance.

Task Requirements

Create a new Jupyter Notebook or Python script.

1. Load Dataset

- Import dataset (e.g., `loans.csv`) using Pandas.

- Display first 10 rows.
 - Check dataset shape and column types.
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2. Data Preprocessing

Perform the following steps:

- Handle missing values (fill or drop).
 - Encode categorical variables (Label Encoding / One-Hot Encoding).
 - Normalize numerical features if required (StandardScaler or MinMaxScaler).
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3. Train-Test Split

- Split dataset into training and testing sets (80/20 ratio).
 - Ensure reproducibility using random state.
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4. Model Building

A. Classification Model

- Use **Logistic Regression**
- Predict: Loan Approval (Approve / Reject)

B. Regression Model

- Use **Linear Regression**
 - Predict: Loan Amount
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5. Model Evaluation

Classification Metrics:

- Accuracy
- Precision
- Recall
- Confusion Matrix

Regression Metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

6. Data Visualization

Create the following visual outputs:

- Confusion matrix heatmap
- Regression line (actual vs predicted values)
- Feature importance (optional)

Submission Requirements

Take screenshots of:

- Data preprocessing steps
- Model training code
- Evaluation results
- All visualizations

Submission Guidelines

Upload your **.docx file** containing code, screenshots, explanations, and results via LMS.